English

(1001CMD303029240044)

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRA...

(Academic Session : 2024-2025)

Test Pattern

MAJOR

23-11-2024

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE - MEA,PS,D,L,M,B,C,N,O,P & MEQ

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

Test Booklet Code

E1

This Booklet contains **28** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology). 50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720**.
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

Your Target is to secure Good Rank in Pre-Medical 2025

SECTION-A (PHYSICS)

1. If force = $\frac{\alpha}{\beta + \text{density}}$, then dimension of $[\alpha\beta]$ is :-

(1) $M^3L^{-3}T^{-3}$

(2) $M^3L^{-2}T^{-3}$

(3) $M^3L^{-5}T^{-2}$

(4) $M^2L^{-3}T^{-2}$

2. Match **List-I** with **List-II** :

	List-I (Measured values)		**List-II (Significant figures)**
(A)	0.001213	I.	2
(B)	2.1×10^{16}	II.	3
(C)	3.70	III.	1
(D)	3000	IV.	4

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-I, B-II, C-IV, D-III

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-II, C-I, D-IV

3. A vernier calipers has 1 mm marks on the main scale. It has 20 equal divisions on the vernier scale which match with 16 main scale divisions. The least count of this vernier calipers is :

(1) 0.05 mm

(2) 0.10 mm

(3) 0.20 mm

(4) 0.02 mm

4. The pitch of a screw gauge is 0.5 mm and there are 50 divisions on its circular scale. When screw is closed and plane face of screw is in contact with plane face of stud, on the frame, it was observed that zero of the circular scale is above the pitch line by 5 divisions. The zero error in instrument is

(1) 0.025mm

(2) −0.25mm

(3) 0.05mm

(4) −0.05mm

5. If the unit of force is 1 kN, unit of length is 1km and the unit of time is 100 s in a new system, then the new unit of mass is :

(1) 1000 kg (2) 1 kg

(3) 10000 kg (4) 100 kg

6. Percentage error in measuring the mass (m) & radius (R) of a solid sphere are 3% & 2% respectively. Then error in measurement of moment of inertia about its diameter will be :

(1) 5% (2) 7%

(3) 4% (4) 8%

7. If energy (E), velocity (v) and force (F), be taken as fundamental quantities, then what are the dimensions of mass?

(1) EV^2 (2) EV^{-2}

(3) FV^{-1} (4) FV^{-2}

8. In the relation $y = r\sin(\omega t - kx)$, the dimensions of ω/k are

(1) $[M^0L^1T^0]$

(2) $[M^0L^0T^1]$

(3) $[M^0L^0T^0]$

(4) $[M^0L^1T^{-1}]$

9. Given below are two statements : one is labelled as Assertion(A) and the other is labelled as Reason (R).

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.

Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

10. One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 10 divisions of vernier scale coincide with 8 small divisions of the main scale, the least count of the callipers is

(1) 0.01 cm

(2) 0.02 cm

(3) 0.05 cm

(4) 0.005 cm

11. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 6 mm and zero of circular scale division is 26 division above the reference level of screw gauge and it has a zero error of 0.003 cm, the correct diameter of the ball is :-

(1) 0.623 cm (2) 0.626 cm

(3) 0.629 cm (4) 0.633 cm

12. In an experiment to find out thickness of a sheet using screw gauge, the following observation were noted :

(a) Screw moves 1 mm on main scale in two complete rotations.

(b) Total divisions on circular scale = 50

(c) Main scale reading is 1.5 mm

(d) 35th division on circular scale coincide with pitch line.

(e) Instrument has negative zero error of 0.02 mm.

Thickness of plate will be :

(1) 0.67 mm

(2) 0.87 mm

(3) 0.53 mm

(4) 1.87 mm

13. The method of dimensional analysis can be used to derive which of the following relation ? .

(1) A_0e^{-kt}

(2) $A\cos(\omega t - kx)$

(3) $ut + \frac{1}{2}at^2$

(4) None of these

14. The length, breadth and thickness of a block are measured as 205.4 cm, 7.0 cm and 0.43 cm respectively. Which one of the following measurements is most accurate measurement ?

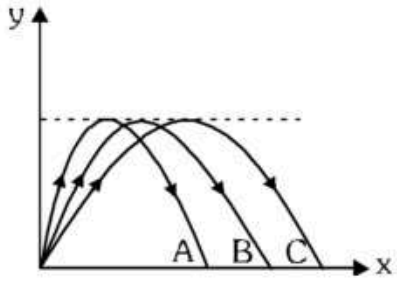
(1) Length

(2) Breadth

(3) Thickness

(4) None of these

15. The trajectories of the motion of three particles A, B and C are shown in figure. Match the entries of column I with the entries of column II.



	Column-I		Column-II
(A)	Time of flight is least for	(P)	A
(B)	Vertical component of the velocity is greatest for	(Q)	B
(C)	Horizontal component of the velocity is greatest for	(R)	C
(D)	Launch speed is least for	(S)	same for all

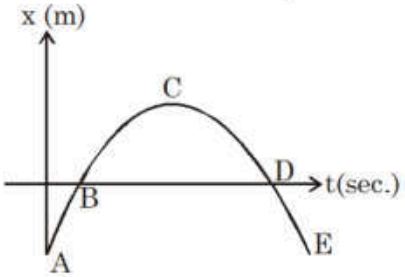
(1) A⟶P, B⟶R, C⟶S, D⟶S

(2) A⟶S, B⟶P, C⟶Q, D⟶R

(3) A⟶S, B⟶S, C⟶P, D⟶R

(4) A⟶S, B⟶S, C⟶R, D⟶P

16. A particle is moving on x-axis. Position-time graph for the particle is as shown in the figure. Select the incorrect statement for motion of particle.



(1) In region CE, particle is moving in negative x-direction with increasing speed.

(2) In region AC, particle is moving in positive x-direction with decreasing speed.

(3) Initial velocity of particle is zero.

(4) Acceleration of particle is in negative x-direction.

17. A person is sitting in a moving train and is facing the engine. He tosses, up a coin which falls behind him. He concludes that the train is moving

(A) forward with increasing speed.

(B) forward with decreasing speed.

(C) backward with increasing speed.

(D) backward with decreasing speed.

Select the alternative :-

(1) Only (A)

(2) Only (A) and (C)

(3) Only (A) and (D)

(4) Only (B) and (C)

18. A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β to come to rest. If the total time elapsed is t, the maximum velocity acquired by the car is given by

(1) $\left[\frac{\alpha\beta}{\alpha+\beta}\right]t$

(2) $\left[\frac{\alpha+\beta}{\alpha\beta}\right]t$

(3) $\left[\frac{\alpha^2+\beta^2}{\alpha\beta}\right]t$

(4) $\left[\frac{\alpha^2-\beta^2}{\alpha\beta}\right]t$

19. A particle is projected with speed 40 m/s at an angle of 53° from horizon from ground. After what time velocity of particle will be perpendicular to the initial velocity ?

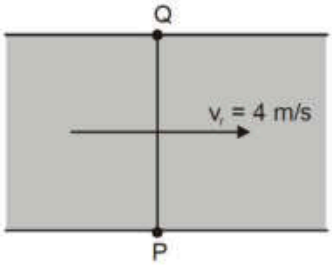
(1) 5 s (2) 10 s (3) 15 s (4) 20 s

20. A body is projected upwards from a height 20 m with speed 25 m/s at an angle 37° from horizontal, then find after how much time it will reach on ground ?

(1) 2 sec (2) 5 sec

(3) 4 sec (4) 1 sec

21. A boat man could row his boat with a speed 10 m/sec. He wants to take his boat from P to a point Q just opposite on the other bank of the river flowing at a speed 4 m/sec. He should row his boat-



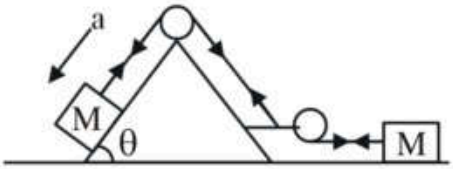
(1) At right angle to the stream

(2) At an angle of $\sin^{-1}\left(\frac{2}{5}\right)$ with PQ up the stream

(3) At an angle of $\sin^{-1}\left(\frac{2}{5}\right)$ with PQ down the stream

(4) At an angle of $\cos^{-1}\left(\frac{2}{5}\right)$ with PQ down the stream

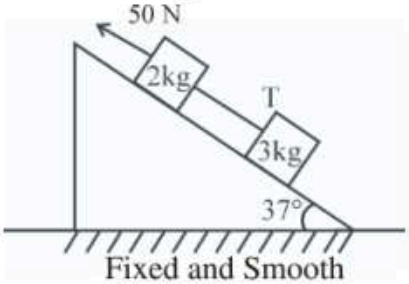
22. Two blocks each having a mass M are moving as shown in the figure. The acceleration of the system is :-



(1) 0 (2) $\frac{g \sin\theta}{2}$

(3) $g \sin\theta$ (4) $2g \sin\theta$

23. The tension in the string is :-



(1) 40N (2) 30N

(3) 20N (4) 28N

24. A diwali rocket is ejecting 0.05 Kg of gases per second at a velocity of 400 m/s. The thrust force on the rocket is :-

(1) 10 N (2) 20 N

(3) 50 N (4) 100 N

25. **Assertion :** The slope of momentum versus time curve give us the acceleration.

Reason : Acceleration is given by the rate of change of momentum.

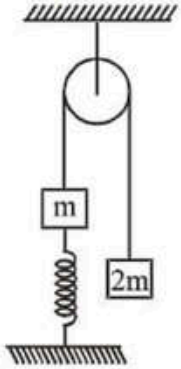
(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are false.

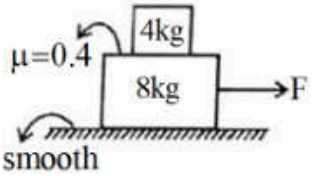
26. A block of mass 2m hangs by an inextensible light string that passes over a smooth pulley. Another mass m hangs from the other end of the string and is connected to a vertical massless spring. The system is stationary. Just after the string is cut, the accelerations of m and 2m, respectively, are :



(1) 0, 0 (2) 0, g ↓

(3) $\frac{g}{2}$ ↓, g ↓ (4) 2g ↓, g ↓

27. Find the maximum force for which both the block will move with same acceleration.



(1) 16 N (2) 32 N (3) 48 N (4) 8 N

28. A rope lies on a table such that some of its part suspended from the table. The rope begins to slide when the length of hanging part is 25 % of entire length. The co-efficient of friction between rope and table is:

(1) 0.33 (2) 0.25 (3) 0.5 (4) 0.2

29. Force of value $30(\hat{i}+\hat{j}+\hat{k})$N displaces a particle from (2, 4, 1)m to (3, 5, 2)m, find work done :

(1) 90 J (2) 95 J

(3) 85 J (4) 100 J

30. A spring of force constant 800 N/m has an extension of 5 cm. The workdone in extending it from 5 cm to 15 cm is :-

(1) 16 J (2) 8 J (3) 32 J (4) 24 J

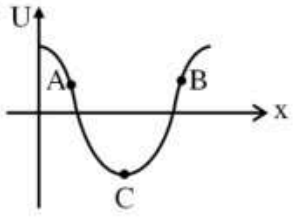
31. An object falls from a height of 10 m above the ground. After striking the ground it loses 50% of its kinetic energy. The height upto which the object can rebounce from the ground is :

(1) 7.5 m (2) 10 m (3) 2.5 m (4) 5 m

32. A stone projected vertically up with a velocity u reaches a maximum height h. When it is at a height of $\frac{2h}{5}$ from the ground, the ratio of KE and PE at that point is : (consider PE = 0 at the point of projection)

(1) $\frac{2}{3}$ (2) $\frac{2}{5}$ (3) $\frac{3}{5}$ (4) $\frac{3}{2}$

33. Potential energy v/s displacement curve for one dimensional conservative field is shown. Force at A, B and C is respectively.



(1) Positive, Positive, Zero

(2) Positive, Negative, Zero

(3) Negative, Positive, Positive

(4) Negative, Negative, Positive

34. An open knife edge of mass 'm' is dropped from a height 'h' on a wooden floor. If the blade penetrates upto depth 'd' into the wood, the average resistance offered by the wood to the knife edge is

(1) mg (2) $mg\left(1-\frac{h}{d}\right)$

(3) $mg\left(1+\frac{h}{d}\right)$ (4) $mg\left(1+\frac{h}{d}\right)^2$

35. A body is initially at rest. It undergoes one-dimensional motion with constant acceleration. The power delivered to it at time t is proportional to-

(1) $t^{1/2}$ (2) t (3) $t^{3/2}$ (4) t^2

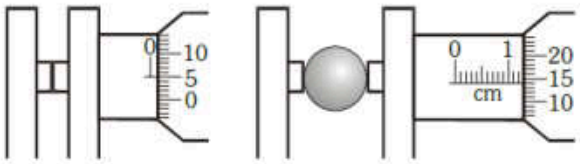
SECTION-B (PHYSICS)

36. In an electrical circuit, the voltage is measured as V = (200±4) volt and the current is measured as I = (20±0.2)A. The value of the resistance is :

(1) (10±4.2)Ω (2) (10±0.3)Ω

(3) (10±0.1)Ω (4) (10±0.1)Ω

37. What is the diameter of sphere shown in figure. Pitch of screwgauge is 1 mm and number of divisions in circular scale are 50 :-



(1) 1.238 cm (2) 1.218 cm

(3) 1.228 cm (4) 1.215 cm

38. The units of length is halved, that of the velocity is doubled and that of the force is trippled. Which of the following is the correct change in the other units ?

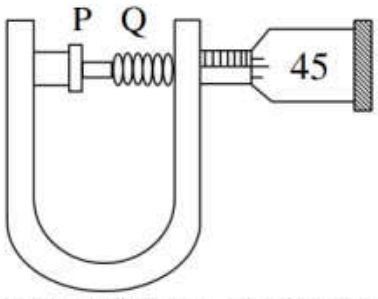
(1) unit of time is doubled

(2) unit of mass is doubled

(3) unit of momentum becomes three fourth

(4) unit of energy is doubled

39. In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(A) Screw moves 0.5 mm on main scale in one complete rotation

(B) Total divisions on circular scale = 50

(C) Main scale reading is 2.5 mm

(D) 45th division of circular scale is in the reference line

(E) Instrument has 0.03 mm negative zero error

Then the diameter of wire is :

(1) 2.92 mm (2) 2.54 mm

(3) 2.98 mm (4) 3.45 mm

40. Length of rectangular box is measured using vernier callipers. If zero of vernier scale lies between 5.1 cm and 5.3 cm reading of main scale and 20 division of vernier scale coincides with 19 divisions of main scale. While measuring length it has been observed that 6th division of vernier scale coincides with main scale division, then length of box will be -

(1) 5.16 mm (2) 5.16 cm

(3) 5.36 cm (4) 5.36 mm

41. The main scale of a vernier calliper has n divisions/cm. (n + 1) divisions of the vernier scale coincide with n divisions of main scale. The least count of the vernier calliper is,

(1) $\frac{1}{(n+1)(n-1)}$ cm

(2) $\frac{1}{n}$ cm

(3) $\frac{1}{n^2}$ cm

(4) $\frac{1}{n(n+1)}$ cm

42. A point particle moves in xy - plane according to the law x = a sin 2t and y = a − a cos 2t where a is a positive constant and t is the time. Match the suitable matching.

List I		**List II**	
(P)	x component of velocity	(1)	constant and equal to 2a
(Q)	y-component of acceleration	(2)	constant and equal to 4a
(R)	Speed	(3)	vary as sinusidally
(S)	Acceleration	(4)	vary as linearly

Codes :-

	P	**Q**	**R**	**S**
(1)	3	3	1	2
(2)	3	4	1	2
(3)	3	2	1	4
(4)	3	4	2	1

43. Water is dripping from the tap at a height h from the ground at regular interval t_1. They reach the ground at interval t_2. Then t_2 is equal to

(1) t_1 (2) $t_1 + \sqrt{2h/g}$

(3) $t_1 - \sqrt{2h/g}$ (4) $t_1 + 2\sqrt{2h/g}$

44. Rain is falling vertically downward with a speed of 50 m/s. Wind starts blowing after some time with a speed of 25 m/s in East to West direction. The direction in which a boy standing at that place should hold his umbrella is:

(1) $\tan^{-1}\left(\frac{1}{3}\right)$ w. r. t. wind

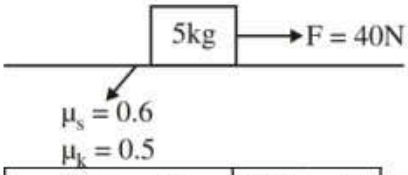
(2) $\tan^{-1}\left(\frac{1}{2}\right)$ w. r. t. rain

(3) $\tan^{-1}\left(\frac{1}{2}\right)$ w. r. t. wind

(4) $\tan^{-1}\left(\frac{1}{3}\right)$ w. r. t. rain

Join Telegram: @NEETvNOAH

45.



Column-I		Column-II	
(P)	a(m/s^2)	(1)	25
(Q)	N (newton)	(2)	30
(R)	f_L(newton)	(3)	50
(S)	f_K (newton)	(4)	3

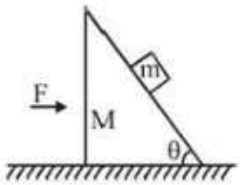
(1) P ⟶ 3, Q ⟶ 4, R ⟶ 2, S ⟶ 1

(2) P ⟶ 4, Q ⟶ 3, R ⟶ 2, S ⟶ 1

(3) P ⟶ 1, Q ⟶ 2, R ⟶ 3, S ⟶ 4

(4) P ⟶ 4, Q ⟶ 2, R ⟶ 1, S ⟶ 3

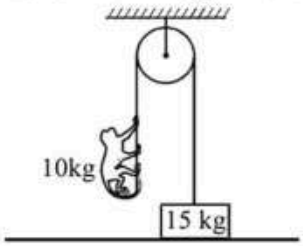
46. A block is kept on a frictionless inclined surface with angle of inclination θ. Value of applied force F for which the block remains at rest with respect to incline -



(1) $\frac{(M+m)g}{\tan\theta}$ (2) $\frac{(M+m)g}{\cos\theta}$

(3) (M + m)gtanθ (4) Mgtanθ

47. A 10kg monkey is climbing a massless rope attached to a 15kg mass and is passing over a smooth pulley. The mass is lying on the ground. In order to raise the mass from the ground he must climb with (g = 10m/s^2)



(1) uniform acceleration greater than 5m/sec^2.

(2) uniform acceleration greater than 2.5m/sec^2.

(3) high speed

(4) uniform acceleration greater than 10m/sec^2 only.

48. upto height of 2 m in 3 sec. The average power generated by it will be :- (g = 9.8 m/s^2)

(1) 1960 W

(2) 2205 W

(3) 4410 W

(4) 0 W

49. **Statement-1 :** A person walking on a horizontal road with a load on his head does no work on the load against gravity.

Statement-2 : No work is said to be done, if directions of force and displacement of load are perpendicular to each other.

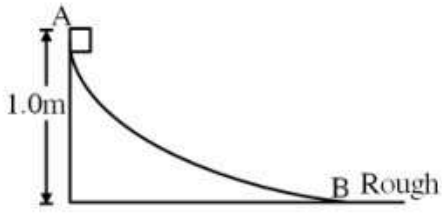
(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

50. A block weighing 20N travels down a smooth curved track AB joined to a rough horizontal surface (figure). The rough surface has a friction coefficient of 0.40 with the block. If the block starts slipping on the track from a point 1.0 m above the horizontal surface, the distance it will move on the rough surface is :



(1) 5.0 m

(2) 2.5 m

(3) 10 m

(4) 7.5 m

SECTION-A (CHEMISTRY)

51. How much oxygen is required for complete combustion of 560 g of ethene ?

(1) 6.4 kg (2) 1.92 kg

(3) 2.8 kg (4) 9.6 kg

52. On analysis, a certain compound was found to contain iodine and oxygen in the mass ratio of 254:80. The formula of the compound is :
(At mass I = 127, O = 16)

(1) I_2O_3 (2) I_2O_4 (3) I_4O_5 (4) I_2O_5

53. At 27°C, one mole gas is allowed to expand reversibly and adiabatically from 10 L to 20 L then change in entropy is :-

(1) Zero (2) +2.5 cal/k

(3) 1.38 cal/k (4) 5.72 cal/k

54. A gas expands from 2L to 6L against a constant pressure of 0.5 atm on absorbing 200 J of heat (1L atm = 101.3 J). Then calculate the change in internal energy :

(1) −2.6 J (2) +2.6 J

(3) −402.6 J (4) +402.6 J

55. If the internal energy of an ideal gas decreases by the same amount as the work done by the gas, then process is :-

(1) cyclic (2) isothermal

(3) adiabatic (4) isochoric

56. For combustion of 1 mole of benzene at 27°C, the heat of reaction at constant pressure is −780.9 kcal. What will be the heat of reaction at constant volume?

$$C_6H_{6(\ell)} + \frac{15}{2}O_{2(g)} \rightarrow 6CO_{2(g)} + 3H_2O_{(\ell)}$$

(1) −781.8 kcal (2) −780.0 kcal

(3) +781.8 kcal (4) +780.0 kcal

57. What will be entropy change for freezing of 10 g water? If enthalpy of fusion of $H_2O(s)$ is 80 cal/g and melting point of ice is 273 K at 1 atm :-

(1) 12.25 cal/K

(2) 2.93 cal/K

(3) −2.93 cal/K

(4) −12.25 cal/K

58. Which of the following equations represents a reaction that provides the heat of formation of CH_3Cl (g)-

(1) $C(s) + HCl(g) + H_2(g) \longrightarrow CH_3Cl(g)$

(2) $C(s) + 3/2H_2(g) + 1/2Cl_2(g) \longrightarrow CH_3Cl(g)$

(3) $C(s) + 3H(g) + Cl(g) \longrightarrow CH_3Cl(g)$

(4) $CH_4(g) + Cl_2(g) \longrightarrow CH_3Cl(g) + HCl(g)$

59. If ΔH_f^o for $H_2O_2(\ell)$ and $H_2O(\ell)$ are −188 kJ mol^{-1} and −286 kJ mol^{-1}, what will be the enthalpy change of the reaction?
$2H_2O_2(\ell) \longrightarrow 2H_2O(\ell) + O_2(g)$

(1) 146 kJ mol^{-1} (2) −196 kJ mol^{-1}

(3) −494 kJ mol^{-1} (4) −98 kJ mol^{-1}

60. The heat of combustion of carbon to $CO_2(g)$ is −393.5 kJ mol^{-1}. Calculate the heat released upon the formation of 35.2 g of $CO_2(g)$ from carbon and oxygen gas :-

(1) 205 kJ (2) 115 kJ

(3) 315 kJ (4) 425 kJ

61. The enthalpy of formation of ammonia is −46.0 kJ mol^{-1}. Then enthalpy change for the reaction is -
$2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

(1) 46.0 kJ mol^{-1}

(2) 92.0 kJ mol^{-1}

(3) −23.0 kJ mol^{-1}

(4) −92.0 kJ mol^{-1}

62. Calculate enthalpy change of the following reaction

$CH_2CH_2\,(g) + H_{2(g)} \longrightarrow CH_3—CH_3\,(g)$

The bond energy of C—H, C—C, C=C, H—H are 414, 347, 615 and 435 kJ mol^{-1} respectively :-

(1) −250 kJ

(2) +250 kJ

(3) +125 kJ

(4) −125 kJ

63. The oxidation state of sulphur in $H_2S_2O_8$ is:-

(1) +2 (2) +4 (3) +6 (4) +7

64. In the reaction ; $8Al + 3Fe_3O_4 \longrightarrow 4Al_2O_3 + 9\,Fe$, the number of electrons transferred from reductant to oxidant is :

(1) 7 (2) 24 (3) 8 (4) 4

65. For the manufacture of ammonia by the reactions, $N_2 + 3H_2 \rightleftharpoons 2NH_3 + 21.9kCal$, the favourable conditions are :

(1) Low temperature, low pressure & catalyst

(2) Low temperature, high pressure & catalyst

(3) High temperature, low pressure & catalyst

(4) High temperature, high pressure & catalyst

66. For the reaction

$CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$

Which one is correct relation :-

(1) $K_p = P^2_{H_2O}$

(2) $K_c = [H_2O]^2$

(3) $K_p = K_c(RT)^2$

(4) All

67. AB dissociates as per reaction

$2AB\,(g) \rightleftharpoons 2A\,(g) + B_2\,(g)$. Initial pressure of AB was 500 mm. of Hg, at equilibrium total pressure becomes 625 mm of Hg. Then K_P for the reaction is (assuming volume remains constant)-

(1) 500 (2) 125 (3) 750 (4) 375

68. The solubility of $PbSO_4$ in 0.01 M Na_2SO_4 solution is (K_{sp} of $PbSO_4 = 1.25 \times 10^{-9}$) :-

(1) 1.25×10^{-7} mol/L

(2) 1.25×10^{-9} mol/L

(3) 1.25×10^{-10} mol/L

(4) 1.25×10^{-18} mol/L

69. What will be the hydrogen ion concentration (moles L^{-1}) in 0.01 M HCN (aq.) solution, if HCN is 20% ionised.

(1) 0.002 (2) 0.02 (3) 0.2 (4) 0.01

70. Match List-I with List-II

List-I		List-II	
(A)	$3Mg(s) + N_2(g) \longrightarrow Mg_3N_2(s)$	(P)	Metal displacement redox reaction
(B)	$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$	(Q)	Decomposition redox reaction
(C)	$2KClO_3(s) \longrightarrow 2KCl(s) + 3O_2(g)$	(R)	Not a redox reaction
(D)	$Zn(s) + CuSO_4(aq.) \longrightarrow ZnSO_4(aq.) + Cu(s)$	(S)	Combination redox reaction

Select the correct answer from options given below :

(1) A-S, B-Q, C-R, D-P

(2) A-S, B-R, C-Q, D-P

(3) A-S, B-P, C-Q, D-R

(4) A-S, B-R, C-P, D-Q

71. For which reaction, relation $K_C = K_P(RT)^{-1}$ is correct.

(1) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(2) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(3) $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$

(4) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

72. Select the incorrect statement-

(1) Reactant which gets consumed and limit the amount of product formed is limiting reagent

(2) An empirical formula represent the correct number of different types of atoms present in a molecule of compound.

(3) Mass of one mole of substance in gram is called its molar mass

(4) Formula mass of sodium chloride is 58.5 u.

73. For reaction, $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ equilibrium concentration of SO_2, O_2 and SO_3 are 0.1 M, 0.2 M and 0.1 M respectively. If volume of vessel is now increased 4 times, then new value of K_c will be-

(1) 0.05 (2) 0.2 (3) 0.1 (4) 0.8

74. Which of the following is true regarding equilibrium constant-

(A) Value of equilibrium constant is independent of initial concentration of the reactants and products.

(B) Equilibrium constant of reverse reaction is equal to the inverse of the equilibrium constant for the forward reaction.

(C) Expression of equilibrium constant is applicable only when concentration of the reactants and products have attained constant value at equilibrium state.

(D) Value of equilibrium constant (K) and reaction quotient (Q) cannot be used to predict the direction of reaction

(E) Equilibrium constant is temperature dependent having one unique value for a particular reaction represented by a balanced equation

Choose the correct answer from the options given below :

(1) A and C only (2) D only

(3) A, B, C and E (4) A, B, C, D and E

75. Dissociation constant of a weak acid HX is 10^{-4}, then pH of 0.01 M solution of its sodium salt is -

(1) 3 (2) 8 (3) 10 (4) 13

76. Which of the following reaction shift in forward direction upon increasing the pressure at equilibrium :-

(1) $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$

(2) $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$

(3) $2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g)$

(4) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

77. Urea (NH_2CONH_2) is a nitrogen fertilizer that is manufactured from NH_3 and CO_2 as per reaction

$2NH_3 + CO_2 \longrightarrow NH_2CONH_2 + H_2O$

if 176 g CO_2 and 102 g NH_3 are reacted then mass of urea produced is :-

(1) 240 g (2) 180 g

(3) 60 g (4) 278 g

78. 200 ml of 0.1 M CH_3COOK (aq.) and 100 ml of 0.1 M HCl(aq.) solutions are mixed, then pH of resulting solution is -

($K_a(CH_3COOH) = 2 \times 10^{-5}$)

(1) 5 (2) 4.7 (3) 10.3 (4) 9

79. In which of the following species underlined element not have +3 oxidation number-

(1) $Na\underline{B}H_4$ (2) $\underline{Fe}_2O_3$

(3) $NaH_2\underline{P}O_3$ (4) $\underline{Pb}Cl_2$

80. In which of the following species metal is not present in fractional oxidation number :-

(1) Fe_3O_4 (2) Mn_3O_4

(3) Pb_3O_4 (4) Fe_2O_3

81. Which of the following species does not show disproportionation -

(1) ClO^- (2) ClO_2^-

(3) ClO_3^- (4) ClO_4^-

82. Solubility product of a salt having formula MX_2 in water is 4×10^{-12}, then concentration of X^- ions in the saturated aqueous solution of the salt is-

(1) 1×10^{-4} M (2) 2×10^{-4} M

(3) 4×10^{-4} M (4) 2×10^{-6} M

83. 10^{-3} moles of KOH is dissolved in 1L water to prepare an aqueous solution, then pH of this solution at 25°C is-

(1) 5 (2) 3 (3) 9 (4) 11

84. For the homogeneous reaction :

$xA + yB \longrightarrow \ell Y + mZ$

$\Delta H = -30$ kJ mol^{-1}, $\Delta S = -100$ J K^{-1} mol^{-1}.

At what temperature the reaction is at equilibrium?

(1) 50°C (2) 250°C

(3) 100 K (4) 27°C

85. One mole of an ideal gas at 300 K is expanded isothermally from 1 L to 10 L volume. ΔU for this process is (Use R = 8.314 J K^{-1} mol^{-1})

(1) 1260 J (2) 2520 J (3) 5040 J (4) 0 J

SECTION-B (CHEMISTRY)

86. ***Assertion*** :- 22.4 L of He gas is having N_A atoms of He at 1 atm. & 273 K.

Reason :- 22.4 L of any gas at 1 atm. & 273 K is having N_A atoms.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

87. A sample of $CaCO_3$ is 50% pure. On heating 1.12 L of CO_2 (g) (at STP) is obtained. Residue left (assuming non-volatile impurity) after heating is :

(1) 7.8 g (2) 3.8 g

(3) 2.8 g (4) 8.9 g

88. half of its initial volume and simultaneously heated to twice its initial temperature, the change in entropy of gas (ΔS) is:-

(1) $C_{p,m} \ln 2$ (2) $C_{v,m} \ln 2$

(3) $R \ln 2$ (4) $(C_{v,m} - R) \ln 2$

89. For the process,

$H_2O(\ell)$ (1 bar, 373 K) $\longrightarrow$ $H_2O(g)$ (1 bar, 373 K),

the correct set of thermodynamic parameters are -

(1) $\Delta G = 0$, $\Delta S = +ve$

(2) $\Delta G = 0$, $\Delta S = -ve$

(3) $\Delta G = +ve$, $\Delta S = 0$

(4) $\Delta G = -ve$, $\Delta S = 0$

90. Calculate the heat of dimerisation of NO_2 on the basis of following reactions :-

(A) $N_2 + 2O_2 \longrightarrow 2NO_2$, $\Delta H = 67.9$ KJ

(B) $N_2 + 2O_2 \longrightarrow N_2O_4$, $\Delta H = 9.3$ KJ

(1) 58.6 KJ (2) -58.6 KJ

(3) 77.2 KJ (4) -77.2 KJ

91. The enthalpy change for the reaction :

$\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$ is -50 kJ/mol.

If the bond dissociation energy of X–X and X–Y is 380 and 150 kJ mol^{-1} respectively, then bond dissociation energy of Y–Y is :-

(1) 35 kJ mol^{-1}

(2) 70 kJ mol^{-1}

(3) 380 kJ mol^{-1}

(4) 140 kJ mol^{-1}

92. What will be the equivalent weight of H_3PO_3 in the following reaction

(M = molar mass of H_3PO_3)

$4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$

(1) $\frac{M}{6}$ (2) $\frac{M}{2}$ (3) $\frac{M}{12}$ (4) $\frac{2M}{3}$

93. 40 mL of 0.1 M ammonia solution is mixed with 20 mL of 0.1 M HCl. Then pH of resultant mixture is? (pK_b of ammonia is 4.74)

(1) 4.74 (2) 2.26 (3) 9.26 (4) 5.00

94. Match List-I with List-II, where List-I contains reactions and list-II contain type of enthalpy corresponding to those reactions :

List-I		List-II	
(A)	$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(\ell)$	(P)	$\Delta H^\circ_{atomization}$
(B)	$CH_4(g) \rightarrow C(g) + 4H(g)$	(Q)	$\Delta H^\circ_{Solution}$
(C)	$AB(s) \rightarrow A^+(aq.) + B^-(aq)$	(R)	$\Delta H^\circ_{formation}$
(D)	$\frac{1}{2}H_2(g) + \frac{1}{2}Br_2(\ell) \rightarrow HBr(g)$	(S)	$\Delta H^\circ_{Combustion}$

Select the correct answer from options given below.

(1) A-P, B-Q, C-R, D-S (2) A-Q, B-P, C-S, D-R

(3) A-S,B-P, C-Q,D-R (4) A-S,B-P,C-R,D-Q

95. Given below are two statements :

Statement-I : $\Delta U = 0$ for both reversible and irreversible isothermal expansion of ideal gas, but ΔS_{total} is not zero for irreversible process.

Statement-II : Total entropy change for system and surrounding in a spontaneous process is negative.

On the basis of above statements, select correct option :

(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

(3) Statement-I is true and Statement-II is false.

(4) Statement-I is false and Statement-II is true.

96. First and second ionisation constant of H_2S are 9×10^{-8} and 1×10^{-13} respectively, then $[S^{-2}]$ in 0.1M H_2S solution if the solution also has 0.1M HCl –

(1) 9×10^{-21} M (2) 9×10^{-20} M

(3) 9×10^{-8} M (4) 9×10^{-7} M

97. Given below are two statements :

Assertion (A) : In reaction

$2Na(s) + H_2(g) \rightarrow 2NaH(s)$, Na is oxidised by H_2.

Reason (R) : Addition of oxygen/electronegative element to a substance is oxidation.

On the basis of above statements, select the correct option :

(1) Both A and R are false

(2) Both A and R are true

(3) A is true but R is false.

(4) A is false but R is true.

98. For given balanced redox reaction (in acidic medium)

$aCr_2O_7^{-2}(aq.) + bSO_3^{-2}(aq.) + cH^+(aq.) \rightarrow 2aCr^{+3}(aq.) + bSO_4^{2-}(aq.) + \frac{c}{2}H_2O(\ell)$

values of a, b and c are respectively :

(1) 3,1,8 (2) 1,3,8 (3) 2,3,8 (4) 2,6,8

99. In which case, equilibrium mixture has appreciable concentrations of both reactants and products :-

(1) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
$K_C = 4.8 \times 10^{-31}$ at 298 K

(2) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
$K_C = 57$ at 700 K

(3) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$
$K_C = 4 \times 10^{31}$ at 300 K

(4) $3O_2(g) \rightleftharpoons 2O_3(g)$
$K_C = 2 \times 10^{-50}$ at 298 K

100. pH of which solution does not depends upon its concentration

(A) HCN(aq.) (B) NaCl(aq.)

(C) NH_4OH(aq.) (D) CH_3COONH_4(aq.)

(E) KBr (aq.)

Choose the correct answer from options given below :-

(1) Only B (2) B and E only

(3) B, D and E (4) D and E only

SECTION-A (BOTANY)

101. Choose the incorrect match:-

(1) Isogamy (flagellated Gametes) - *Ulothrix*

(2) Isogamy (Non-flagellated gametes) - *Spirogyra*

(3) Anisogamy - *Sargassum*

(4) Oogamy - Chara

102. In which of the following all examples are of members of phaeophyceae ?

(1) *Spirogyra, Volvox, Fucus*

(2) *Laminaria, Volvox, Polysiphonia*

(3) *Fucus, Laminaria, Sargassum*

(4) *Ulothrix, Laminaria, Sargassum*

103. The members of Rhodophyceae are red in colour due to predominance of pigment :-

(1) Fucoxanthin (2) r-Phycoerythrin

(3) Xanthophylls (4) Anthocyanin

104. Which of the following is a correct match of number of flagella and position of its insertion in algae ?

(1) Red algae - 2, equal, lateral

(2) Brown algae - 2, unequal, lateral

(3) Green algae - flagella absent

(4) Blue green algae - 2-8, equal, apical

105. Given below are two statements

Statement I : Bryophytes play an important role in plant succession on bare rocks.

Statement II : Mosses form dense mats on soil.

In the light of the above statements. Choose the correct answer from the options given below.

(1) Both Statement I and statement II are false

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are true

106. In Bryophytes, spore germinate to produce:-

(1) Prothallus

(2) Sporophyte

(3) Gametophyte

(4) Sporangia

107. Branched rhizoids and leafy gametophytes are characteristics of :-

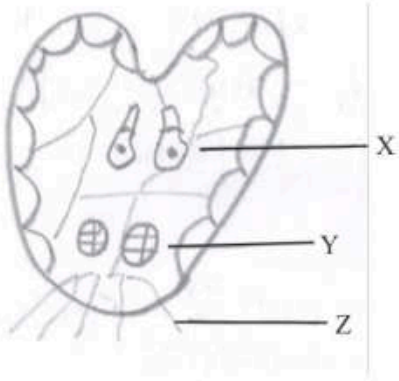
(1) All bryophytes

(2) Mosses

(3) Algae

(4) Liverworts

108. What are X, Y and Z respectively in the diagram below-



(1) Antheridia, Archegonia, Rhizoids

(2) Archegonia, Antheridia, Rhizoids

(3) Antheridia, Archegonia, Seta

(4) Archegonia, Antheridia, Seta

109. Which of the following genera is homosporous and heterosporous respectively.

(1) *Salvinia* & *Selaginella*

(2) *Dryopteris* & *Adiantum*

(3) *Lycopodium* & *Selaginella*

(4) *Equisetum* & *Pteris*

110. Plants of *Pinus* and *Cycas* are respectively

(1) Dioecious, Monoecious

(2) Dioecious, Dioecious

(3) Monoecious, Monoecious

(4) Monoecious, Dioecious

111. Which one of the following is tallest tree species ?

(1) *Cycas* (2) *Cedrus*

(3) *Sequoia* (4) *Pinus*

112. Why the two kingdom classification used for a long time was found inadequate ?

(1) Two kingdom classification was easily done.

(2) Two kingdom classification was easy to understand.

(3) In two kingdom classification simple morphological characters were used.

(4) A large number of organisms did not fall into either categery.

113. **Statement - I :** Archaebacteria are able to survive in harsh habitat.
Statement - II : Archaebacteria survive in extreme condition due to the presence of peptidoglycan in their cell wall.

(1) Statement I is correct and statement II is incorrect.

(2) Statement II is correct and statement I is incorrect.

(3) Both statement I & statement II incorrect.

(4) Both statement I & statement II are correct.

114. Some of bacteria act as pathogen causing ___A___ in animal and ___B___ in plants. Select correct option to fill A & B.

(1) A – Cholera, B – Typhoid

(2) A – Crown gall, B – Citrus canker

(3) A – Tetanus, B – Citrus canker

(4) A – Crown gall, B – Bacterial leaf blight

115. In some prokaryotes like cyanobacteria there are membranous extensions into the cytoplasm which contain pigments, are called :-

(1) Chromatophores

(2) Mesosomes

(3) Ribosomes

(4) Gas vacuoles

116. Read column-I and II and identify their correct match :-

	Column-I		**Column-II**
A.	Thermoacidophiles	(i)	Marine habitat
B.	Dinoflagellate	(ii)	Symbiont with fungi
C.	Algae	(iii)	Guts of ruminants
D.	Methanogens	(iv)	Hot spring

Select the correct answer from the option given below :-

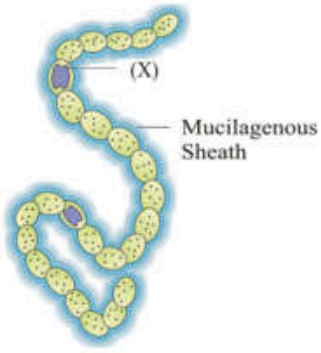
(1) A-(iv), B-(i), C-(ii), D-(iii)

(2) A-(iv), B-(ii), C-(i), D-(iii)

(3) A-(i), B-(ii), C-(iii), D-(iv)

(4) A-(ii), B-(i), C-(iii), D-(iv)

117. What is 'X' in the structure of *Nostoc.*



(1) Plasmodium

(2) Pseudopodia

(3) Heterocyst

(4) Photosynthetic cell

118. ____A____ are organism that completely lack a cell wall. They are the ____B___ living cell known and can survive without oxygen.

(1) A-Archaebacteria, B-Longest

(2) A-Mycoplasma, B-Smallest

(3) A-Eubacteria, B-Smallest

(4) A-Cyanobacteria, B-Longest

119. Archaebacteria differ from other bacteria in having :-

(1) Definite nuclear structure

(2) Different cell wall structure

(3) Some membranous cell organelles

(4) Photosynthetic pigments

120. Read the following statement regarding euglenoids and select the incorrect statements -

(i) They are photosynthetic in the presence of sunlight but become heterotrophs in the absence of sunlight.

(ii) They usually posses two flagella one long and one short.

(iii) Their body is covered by a protein rich layer called pellicle which makes their body thick.

(iv) These are mostly fresh water organisms found in stagnant water.

(v) They are multicellular ciliate protists.

(1) (ii), (iii) and (v)

(2) (i), (iii) and (v)

(3) (iii) and (v)

(4) (iii), (iv) and (v)

121. Match column-I and column-II and select the correct option from the following-

Column-I		Column-II	
I	Diatoms	a	Stiff cellulose plates
II	*Euglena*	b	Thin overlapping shells
III	Slime moulds	c	Chl 'a' & 'b'
IV	Dinoflagellates	d	Plasmodium

(1) I-a, II-d, III-c, IV-b

(2) I-b, II-c, III-d, IV-a

(3) I-b, II-d, III-c, IV-a

(4) I-c, II-b, III-a, IV-d

122. Which of the following organisms is devoid of cell wall but their spores possess true walls ?

(1) *Euglena*

(2) Slime mould

(3) Diatoms

(4) *Gonyaulax*

123. Which of the following statement is incorrect ?

(1) Fungi constitute a unique kingdom of heterotrophic organisms.

(2) The body of fungi consist of long, slender thread-like structure called hyphae.

(3) The cell walls of fungi are composed of chitin and polypeptides.

(4) Fungi prefer to grow in warm and humid places.

124. Which of the following is the correct sequence of steps involves in the sexual cycle of fungi ?

(1) Plasmogamy, meiosis, karyogamy

(2) Meiosis, Karyogamy, plasmogamy

(3) Karyogamy, plasmogamy, meiosis

(4) Plasmogamy, karyogamy, meiosis

125. The imperfect fungi which are decomposer of litter and help in mineral cycling belong to ?

(1) Basidiomycetes (2) Deuteromycetes

(3) Phycomycetes (4) Ascomycetes

126. _____A____ is used extensively in biochemical and genetic work, many members like _____B____ are edible and are considered delicacies.

(1) A-*Claviceps*, B-*Rhizopus* and *Mucor*

(2) A-*Aspergillus*, B-*Agaricus*

(3) A-*Puccinia*, B-*Ustilago*

(4) A-*Neurospora*, B-Morels and truffles

127. Which of the following statement is incorrect regarding class ascomycetes ?

(1) Members of ascomycetes are commonly known as sac-fungi.

(2) In ascomycetes mycelium is branched and septate.

(3) The ascomycetes are mostly unicellular and rarely multicellular.

(4) In ascomycetes asexual reproduction takes place by the formation of conidia.

128. In basidiomycetes, karyogamy and meiosis take place in ?

(1) Basidium

(2) Conidium

(3) Basidiospore

(4) Dikaryon

129. __A__ showed that viruses could be crystallised and crystal consist largely of __B__.

(1) A-Ivanowesky B-DNA or RNA

(2) A-Stanley B-Protein

(3) A-Beijerinek B-Lipid

(4) A-Diener B-RNA

130. Potato spindle Tuber disease is caused by:

(1) Virus

(2) Viroids

(3) Prions

(4) Bacteriophage

131. The fungal partner in a Lichen is called mycobiont whereas the algal partner is called

(1) Glycobiont

(2) Algobiont

(3) Protobiont

(4) Phycobiont

132. How many of the following structures in the list given below are found in both Monerans and Protistans ?

Ribosomes, Plasmid, Well-defined Nucleus, Cell wall, Mesosome, Mitochondria, Flagella, Zygote.

(1) Four (2) Three (3) Five (4) Six

133. The term systematics was derived from latin word "Systema" which means-

(1) Taxonomy of organisms.

(2) Evolutionary study of organisms.

(3) Systematic arrangement of organisms.

(4) Use of various organisms.

134. Which one of the advantage of binomial nomenclature?

(1) Binomial nomenclature vary from place to place.

(2) This naming system providing a name with single component.

(3) This naming system using a single word format was found convenient.

(4) It ensure that each organism has only one name.

135. Taxonomic categories are distinct A and not merely B .

	A	B
(1)	Chemical entities	Morphological aggregates
(2)	Biological entities	Anatomical aggregates
(3)	Chemical entities	Anatomical aggregates
(4)	Biological entities	Morphological aggregates

SECTION-B (BOTANY)

136. **Assertion (A) :** In *Volvox* and *Fucus* oogamous type of sexual reproduction is found.

Reason (R) : In *Volvox* and *Fucus* fusion between one large, non motile (static) female gamete and a smaller motile male gamete occurs.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) (A) is correct but (R) is incorrect

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

137. Usually the members of chlorophyceae are grass green due to dominance of pigments :-

(1) Chlorophyll a & d

(2) Chlorophyll c & d

(3) Chlorophyll b & c

(4) Chlorophyll a & b

138. **Assertion (A) :** Bryophytes are known as amphibians of the plant kingdom.

Reason (R) : Bryophytes live in soil but are dependent on water for sexual reproduction.

(1) Both Assertion & Reason are True and the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

139. Read the following terms carefully and find out how many are related with Gymnosperms?
Nucellus, Antheridia, Archegonia, Flower, Fruit, Endosperm, Coralloid root, Pollen grain, Seed, Ovule, Double fertilisation, Embryo

(1) 6

(2) 7

(3) 9

(4) 8

140. Which of the following kingdoms have multicellular heterotroph organisms ?

(1) Monera, Protista and Animalia only

(2) Protista, Fungi and Animalia only

(3) Fungi and Animalia only

(4) Monera, Protista, Fungi and Animalia only

141. Which of the following structure provides strong structural support to prevent bacterium from bursting or collapsing ?

(1) Cell membrane

(2) Fimbriae

(3) Cell wall

(4) Slime layer

142. Bacteria as a group show the most extensive __A__ diversity so they are very __B__ in behaviour.
Select correct option to fill A & B :-

(1) A-Morphological, B-Simple

(2) A-Nutritional, B-Simple

(3) A-Metabolic, B-Complex

(4) A-Reproductive, B-Complex

143. Which of the following statements are true ?
(a) Vast majority of bacteria are autotrophs.
(b) Bacteria reproduce mainly by fission.
(c) Mycoplasma are pathogenic to plants only.
(d) All cyanobacteria can fix atmospheric nitrogen.
Option :

(1) a, b and c only

(2) a, c and d only

(3) b only

(4) b, c only

144. An 'X' undergo such a rapid multiplication that they make the sea appear red and kills large marine animals like 'Z'. 'X' belongs to 'Y'. Identify 'X', 'Y' and 'Z'.

(1) X = *Paramoecium*, Y = protozoa, Z = Crocodiles

(2) X = *Gonyaulax*, Y = Dinoflagellate, Z = Fishes

(3) X = *Tryponosoma*, Y = Protozoa, Z = Frogs

(4) X = *Gonyaulax*, Y = Diatom, Z = Fishes

145. Which of the following statement is incorrect ?

(1) Chrysophytes are microscopic and float passively in water currents.

(2) Dinoflagellates, walls are embedded with silica and thus the walls are indestructible.

(3) Slime moulds are saprophytic protists.

(4) All protozoans are heterotops and live as predators or parasites.

146. Among Rust, Smut and Mushroom, all the three -

(1) Are pathogens

(2) Are Saprobes

(3) Bear Ascospores

(4) Bear Basidiospores

147. Read the following statements carefully -
(A) Phycomycetes are found in Aquatic habitat.
(B) Ascomycetes are saprophytic, decomposers, parasitic or coprophilous.
(C) Basidiomycetes are grow in soil on logs and tree stumps and in living plants bodies as parasite.
(D) Some members of deuteromycetes are parasites or saprophytes.
How many above statements are correct.

(1) One (2) Two

(3) Three (4) Four

148. M.W. Beijerinek demonstrated the extract of infected plant of __A__ could cause infection in healthy plant and fluid is called as __B__.

(1) A-Soyabean B-Intercellular fluid

(2) A-Pea B-Interstitial fluid

(3) A-Rajma B-Poisonous fluid

(4) A-Tobacco B-Infectious living fluid

149. Mycorrhiza is a symbiotic association between :-

(1) Algae and fungi

(2) Algae and roots of higher plants

(3) Fungi and roots of higher plants

(4) Bacteria and Legume root.

150. In a taxonomic hierarchy, genus is placed between :

(1) Kingdom and class

(2) Class and family

(3) Family and order

(4) Species and family

SECTION-A (ZOOLOGY)

151. In female cockroach, genital pouch contain-

(1) Female gonopore

(2) Spermathecal pore

(3) Collateral gland

(4) All

152. Choose the odd one w.r.t. its number in cockroach.

(1) Labrum (2) Labium

(3) Mandible (4) Hypopharynx

153. On an average female cockroach produces _____ ootheca in her lifetime.

(1) 9 – 10 (2) 14-16 (3) 10-12 (4) 8-10

154. Which one of the following statement is NOT correct regarding anal style in cockroach ?

(1) Short thread like structure

(2) These are one pair

(3) Present in male cockroach

(4) Present in female cockroach

155. Match **List–I** with **List–II**

List–I (Cockroach)		**List–II (Term used)**	
A.	Metamorphosis	I.	Ommatidia
B.	Brain	II.	Paurometabolous
C.	Vision	III.	Spiracles
D.	Respiration	IV.	Supra–oesophageal ganglion

Choose the **correct** answer from the options given below :

(1) A–I, B–III, C–IV, D–II

(2) A–III, B–II, C–I, D–IV

(3) A–II, B–IV, C–I, D–III

(4) A–II, B–I, C–IV, D–III

156. Read the following statements regarding Periplaneta americana ?

(A) Head holds a bit of nervous system while the rest is situated along the ventral (belly - side) part of its body

(B) The next to last nymphal stage has wing pads but only adult cockroach have wings.

(C) Blood enters into heart chambers through ostia during diastolic stage and is pumped anteriorly during systolic stage

(D) Maximum digestion takes place in crop but it get completed in mesenteron.

Which of the above statements are **correct** :

(1) A and B only (2) B, C and D

(3) A, B and D only (4) A, B, C and D

157. From the list of given characters, which of them represent *Chondrichthyes* & *Osteichthyes* ?

(a) Placoid scales

(b) Operculum present

(c) Air bladder present

(d) Mostly viviparous

(e) External fertilisation

(f) Notochord persistent throughout life.

	Chondrichthyes	**Osteichthyes**
(1)	a, b, d	c, e, f
(2)	a, d, f	b, c, e
(3)	b, c, d	a, e, f
(4)	b, c, e	a, d, f

158. A file like rasping organ radula is found in -

(1) Sepia (2) Asterias

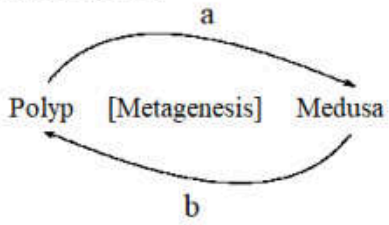
(3) Balanoglossus (4) Musca

159. An animal phylum having radially symmetrical adults but bilateral symmetrical larvae is :

(1) Porifera (2) Coelenterata

(3) Echinodermata (4) Annelida

160. Recoginse the figure and find out the correct option about a and b :



(1) a — sexually, b — asexually

(2) a — asexually, b — sexually

(3) a — asexually, b — asexually

(4) a — sexually, b — sexually

161. Metameric segmentation is present in :

(1) Platyhelminthes, Annelida, Mollusca

(2) Annelida, Arthropoda, Mollusca

(3) Annelida, Arthropoda, Chordata

(4) Enchinodermata, Hemichordata, Chordata

162. Body having meshwork of cells, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum :

(1) Protozoa (2) Coelenterata

(3) Porifera (4) Mollusca

163. Cellular grade of organisation is found in :-

(1) Sponges (2) Coelentrates

(3) Platyhelminthes (4) Ctenophora

164. Choose the true fish among the following.

(1) Dog fish (2) Silver fish

(3) Star fish (4) Devil fish

165. Exclusively marine urochordate among the following is

(1) *Salamandra*

(2) *Ascidia*

(3) *Amphioxus*

(4) *Petromyzon*

166. Identify the character among the following that is shared by *Carcharodon* and *Balaenoptera.*

(1) Ability to regulate body temperature

(2) Notochord is persistent throughout life

(3) Internal fertilisation and viviparity

(4) Two-chambered heart

167. In which of the following animals muscular pharynx is more distinct :-

(1) Wuchereria

(2) Taenia

(3) Loligo

(4) Pila

168. Some animals have body cavity not lined by mesoderm, instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm, the animals can be identified as ?

(1) Tape worms (2) Round worms

(3) Leeches (4) Snails

169. **Assertion :-** Aves and mammals are homeothermal animals.

Reason :- Aves and mammals are able to maintain a constant body temperature.

(1) **Assertion** is correct but **Reason** is incorrect

(2) **Assertion** is incorrect but **Reason** is correct

(3) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**

(4) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**

170. There are some features of animals are given :-
Organ system level, Coelomate, Open circulation, Triploblastic, Unsegmented, Intracellular digestion.
How many characters are present in the members of phylum arthropoda -

(1) 3 (2) 4 (3) 5 (4) 2

171. **Statement-I** : Chondrichthyes skin is tough & have placoid scales.
Statement-II : Air bladder provides buoyancy in cartilaginous fishes.

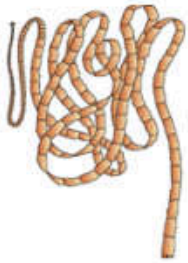
(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

172. Given are the figure of four animals (A),(B),(C) and (D). Select the **incorrect** option with respect to a common characteristic of these animals:

(A)  (B) 

(C)  (D) 

(1) (A)-Tapeworm: False segmentation, triploblastic

(2) (B)-*Aurelia*: Polyp form, stinging cells are present on body.

(3) (C)-*Octopus*: Body is unsegmented

(4) (D)-Scorpion: They have jointed appendages

173. Notochord is a _____ derived rod like structure formed on the _____ side during embryonic development in some animals :-

(1) Mesodermally, dorsal

(2) Ectodermally, dorsal

(3) Mesodermally, ventral

(4) Ectodermally, ventral

174. In a frog brain optic lobes are a part of

(1) Fore brain

(2) Mid brain

(3) Hind brain

(4) Cerebral hemispheres

175. The frog's body is divisible into :-

(1) Head, neck, abdomen

(2) Head, neck, Trunk

(3) Head, Trunk

(4) Head, neck, Trunk & Tail

176. The main excretory product in frog is :-

(1) Ammonia (2) Uric acid

(3) Amino acid (4) Urea

177. Which of the following is correct for circulatory system of heart of frog ?

(1) 3-chambered heart, open circulation, sinus-venosus

(2) 3-chambered heart, close-circulation, 2-ventricles

(3) 3-chambered heart, open-circulation, 2-Atria

(4) 3-chambered heart, close-circulation, sinus-venosus

178. Match column-I with column-II and select correct option :-

Column-I		Column-II	
(a)	Sensory papillae	(i)	Well organised structure
(b)	Nasal epithelium	(ii)	Cellular aggregation
(c)	Eyes	(iii)	Smell
(d)	Taste buds	(iv)	Organs of touch

(1) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)

(2) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(3) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(4) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)

179. The matrix or ground substance in connective tissues are made up of _____

(1) Thick proteins

(2) Elastin fibres

(3) Modified polysaccharides

(4) Modified triglycerides

180. Tendon are example of –

(1) Dense regular connective tissue

(2) Dense irregular connective tissue

(3) Areolar tissue

(4) Adipose tissue

181. The lining of Fallopian tube is made up of :-

(1) Squamous epithelium

(2) Ciliated columnar epithelium

(3) Brush border columnar epithelium

(4) Cuboidal epithelium

182. Specialised connective tissue includes :-
(A) Cartilage (B) Bone (C) Blood (D) Tendon (E) Ligament

(1) Only A,D

(2) Only B,E

(3) Only A, B & C

(4) Only C

183. How many of the following are examples of simple cuboidal epithelium?
Proximal convoluted tubule, Moist surface of Buccal cavity, Urinary bladder, Stomach, Lining of fallopian tube, Ducts of glands

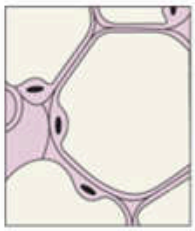
(1) Two

(2) Three

(3) Four

(4) Five

184. How many of the following statements are correct about the given diagram



i - This tissue is located mainly beneath the skin.
ii - The cells of this tissue are specialised to store fats.
iii - This tissue is a type of loose connective tissue.
iv - It serves as a support frame work for epithelial tissue.

(1) 1 (2) 2 (3) 3 (4) 4

185. The epithelial tissue has the following features:
A. Adipose cells are abundant in areolar connective tissue found beneath skin.
B. Provides a lining for some parts of body.
C. Compound epithelium is multilayered meant for limited role in secretion and absorption.
D. Located in skin.
E. Glands are modified epithelial tissue.
Choose the **most appropriate** answer from the options given below:

(1) A, B, C and D

(2) B, C, D and E only

(3) A, C, D and E only

(4) A, B, D and E only

SECTION-B (ZOOLOGY)

186. Which type of mouth parts present in cockroach-

(1) Biting and chewing

(2) Biting and Sucking

(3) Sucking and chewing

(4) Sucking and churning

187. Given below are two statements, choose the correct option :

Statement-I :- In male cockroach, testes present on 2nd-4th abdominal segments.

Statement-II :- In cockroach, the ejaculatory duct opens into male gonopore situated ventral to anus.

(1) Both statement I & II are true

(2) Both statement I & II are false

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

188. Which annelid is dioecious.

(1) Earthworm (2) Leech

(3) Nereis (4) Ascaris

189. Poikilothermous animals with two-chambered heart and internal fertilization belong to taxon

(1) Chondrichthyes

(2) Amphibia

(3) Reptilia

(4) Mammalia

190. Identify the Limbless amphibia.

(1) Hyla (2) Bufo

(3) Salamandra (4) Ichthyophis

191. All animals of which phylum are exclusively marine?

(1) Ctenophora (2) Echinodermata

(3) Mollusca (4) Both 1 and 2

192. Match the List-1 and List-2 :

	List-1		List-2
(i)	Testudo	(a)	Mammalia
(ii)	Aptenodytes	(b)	Reptiles
(iii)	Felis	(c)	Aves

(1) i-a, ii-b, iii-c (2) i-c, ii-b, iii-a

(3) i-b, ii-c, iii-a (4) i-b, ii-a, iii-c

193. Consider the following statements :-

During the alternation of generation or metagenesis in dimorphic cnidarians the polyps are :-

(A) Sessile

(B) Free living

(C) Polyp forms the medusae sexually

(D) Polyp forms the medusae asexually

(1) A, C & D are correct

(2) A and D are correct

(3) B, C & D are correct

(4) A, B, C are correct

194. During hibernation frog respires through-

(1) Skin only

(2) Lungs only

(3) Both

(4) None

195. Which of the following structures in frog directly receives blood through vena cava–

(1) Right atrium

(2) Left atrium

(3) Sinus venosus

(4) Conus arteriosus

196. **Assertion:** Frog also respire through skin.

Reason: Frog have a mucous glands in its skin and skin is highly vasculized.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

197. In multicellular animals, a group of similar cells along with intercellular substances perform a specific function. Such an organisation is called:-

(1) Organ system

(2) Organ

(3) Tissue

(4) Individual

198. Which epithelium is found in the lining of intestine and help in secretion and absorption ?

(1) Ciliated columnar epithelium

(2) Cuboidial epithelium with cilia

(3) Columnar epithelium without microvilli

(4) Columnar epithelium with microvilli

199. Select the correct statement from the following given statements ?

(1) Group of dissimilar cells along with intracellular substance perform a specific function called tissue.

(2) Complex organism like humans consist of only three types of tissue.

(3) Splitting of work exhibit the division of labour.

(4) Brain consist of all three types of tissues.

200. Which of the following statements are **incorrect** ?

(a) Each organ in our body is made of one or more type of tissues.

(b) Cell junction provide only structural links between it's individual cells.

(c) Tight junction help to stop substances from leaking across a tissue.

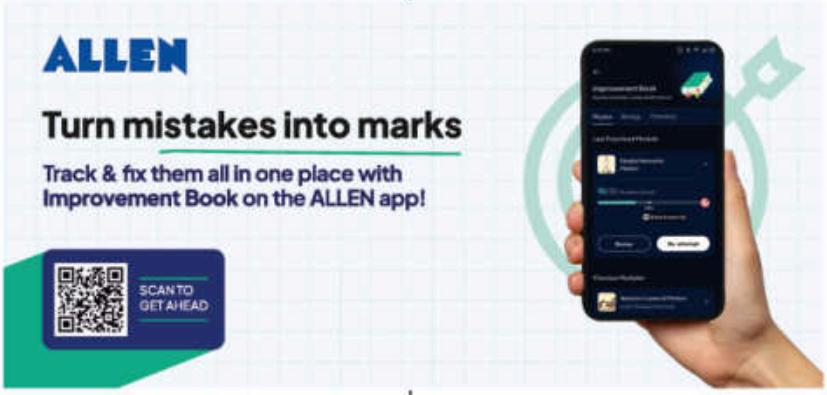
(d) Cuboidal epithelium is found in thin part of loop of henle.

(e) All glands are composed of both epithelial and connective tissue

(1) a, b and c (2) b, c and d

(3) b, d and e (4) a, b, d and e





ENTHUSIAST MAJOR TEST PLAN

SESSION 2024-25

MEA,MEPS,MEB,MEC,MED,MEL,MEM,MEN,MEO,MEP,MEQ

S. NO	DATE	DAY	SYLLABUS
1	23/11/2024	Saturday	SYLLABUS-1
2	28/11/2024	Thursday	SYLLABUS-2
3	03/12/2024	Tuesday	SYLLABUS-3
4	08/12/2024	Sunday	SYLLABUS-4
5	13/12/2024	Friday	SYLLABUS-5
6	18/12/2024	Wednesday	SYLLABUS-6
7	19/12/2024	Thursday	BOARD PATTERN (ENGLISH) (OFFLINE)
8	22/12/2024	Sunday	SYLLABUS-1+2
9	24/12/2024	Tuesday	BOARD PATTERN (CS/PHY.EDU/HINDI) (OFFLINE)
10	27/12/2024	Friday	SYLLABUS-3+4
11	28/12/2024	Saturday	BOARD PATTERN (CHEMISTRY) (OFFLINE)
12	01/01/2025	Wednesday	SYLLABUS-5+6
13	03/01/2025	Friday	BOARD PATTERN (BIOLOGY) (OFFLINE)
14	06/01/2025	Monday	SYLLABUS-1+2+3
15	07/01/2025	Tuesday	BOARD PATTERN (PHYSICS) (OFFLINE)
16	08/01/2025	Wednesday	BOARD PATTERN (CHEMISTRY) (OFFLINE)
17	11/01/2025	Saturday	SYLLABUS-4+5+6
18	12/01/2025	Sunday	BOARD PATTERN (BIOLOGY) (OFFLINE)
19	15/01/2025	Wednesday	FULL SYLLABUS
20	16/01/2025	Thursday	BOARD PATTERN (PHYSICS) (OFFLINE)
21	20/01/2025	Monday	FULL SYLLABUS
22	24/01/2025	Friday	FULL SYLLABUS
23	28/01/2025	Tuesday	FULL SYLLABUS
24	01/02/2025	Saturday	FULL SYLLABUS
25	05/02/2025	Wednesday	FULL SYLLABUS

ENTHUSIAST PROPOSED SYLLABUS 1 to 6 (SESSION 2024-2025)

SUBJECTS	PHYSICS	CHEMISTRY	BIOLOGY
SYLLABUS 1	» Error And Measurement » Kinematics » Laws Of Motion » Work, Energy, And Power » *Experimental Skills* * Vernier Calipers-Its Use To Measure The Internal And External Diameter And Depth Of A Vessel. * Screw Gauge-Its Use To Determine Thickness/ Diameter Of Thin Sheet/Wire.	» Some Basic Concepts of Chemistry » Equilibrium » Chemical Thermodynamics » Redox reactions	» The Living World » Biological Classification » Plant Kingdom » Structural Organisation In Animals (Animal Tissue) » Animal Kingdom » Cockroach, Frog
SYLLABUS 2	» Center Of Mass And Collision » Circular Motion » Rotational Motion » Gravitation » Properties Of Solids And Liquids » *Experimental Skills* * Young'S Modulus Of Elasticity Of The Material Of A Metallic Wire. * Surface Tension Of Water By Capillary Rise And Effect Of Detergents. * Co-Efficient Of Viscosity Of A Given Viscous Liquid By Measuring Terminal Velocity Of A Given Spherical Body.	» Atomic Structure » Classification of Elements and Periodicity in Properties » Chemical Bonding & Molecular Structure	» Cell : The Unit Of Life » Cell Cycle And Cell Division » Biomolecules, Enzyme » Excretory Products And Their Elimination (Excretory System) » Breathing And Exchange Of Gases (Respiratory System) » Body Fluids And Circulation (Circulatory System)
SYLLABUS 3	» Thermodynamics » Kinetic Theory Of Gases » Oscillations And Waves » *Experimental Skills* * Simple Pendulum-Dissipation Of Energy By Plotting A Graph Between The Square Of Amplitude And Time. * Metre Scale - The Mass Of A Given Object By The Principle Of Moments. * Speed Of Sound In Air At Room Temperature Using A Resonance Tube. * Specific Heat Capacity Of A Given (I) Solid And (II) Liquid By Method Of Mixtures.	» Some Basic Principles of Organic Chemistry » Hydrocarbons » Purification and Characterisation of Organic Compounds.	» Photosynthesis in Higher Plants » Plant Growth and Development » Respiration in Plants » Neural Control And Co-ordination (Nervous System) » Chemical Co-ordination And Integration (Endocrine System) » Locomotion And Movement (Muscles, Skeletal System)
SYLLABUS 4	» Electrostatics & Capacitor » Current Electricity » *Experimental Skills* * The Resistivity Of The Material Of A Given Wire Using A Metre Bridge. * The Resistance Of A Given Wire Using Ohm'S Law. * Resistance And Figure Of Merit Of A Galvanometer By Half Deflection Method.	» Chemical Kinetics » Solutions » Electrochemistry » *Principles Related to Practical Chemistry :-* The chemistry involved in the titrimetric exercises-Acids, Bases and the use of indicators, oxalic acid vs $KMnO_4$, Mohr's salt vs $KMnO_4$ Chemical principles involved in the following experiments. 1. Enthalpy of solution of $CuSO_4$. 2. Enthalpy of neutralization of strong acid and strong base. 3. Preparation of lyophilic and lyophobic sols. 4. Kinetic study of the reaction of iodide ions with hydrogen Peroxide at room temperature.	» Sexual Reproduction in Flowering Plants » Morphology Of Flowering Plants » Anatomy Of Flowering Plants » Human Reproduction » Reproductive Health
SYLLABUS 5	» Magnetic Effects Of Current And Magnetism » Electromagnetic Induction And Alternating Currents » Electromagnetic Waves	» p-block elements » d & f-block elements » Coordination Compounds » *Principles Related to Practical Chemistry :-* 1. The chemistry involved in the preparation of Inorganic compound; Mohr's salt, potash alum. 2. Chemical principles involved in the qualitative salt analysis. 3. Cation - Pb^{2+}, Cu^{+2}, Al^{3+}, Fe^{3+}, Zn^{+2}, Ni^{2+}, Ca^{2+}, Ba^{2+}, Mg^{2+}, NH_4^+ 4. Anions - CO_3^{2-}, S^{2-}, SO_4^{2-}, NO_2^-, Cl^-, Br^-, I^- (Insoluble Salt Excluded)	» Principles Of Inheritance And Variation » Molecular Basis Of Inheritance, Microbes In Human Welfare » Biotechnology : Principles And Processes » Biotechnology And Its Applications
SYLLABUS 6	» Optics » Dual Nature Of Matter And Radiation » Atoms And Nuclei » Electronic Devices » *Experimental Skills* * The Focal Length Of; (I)-Convex Mirror (II)-Concave Mirror (III) Convex Lens, Using The Parallax Method. * The Plot Of The Angle Of Deviation Vs Angle Of Incidence For A Triangular Prism. * Refractive Index Of A Glass Slab Using A Travelling Microscope. * Characteristic Curves Of A P-N Junction Diode In Forward And Reverse Bias. * Characteristic Curves Of A Zener Diode And Finding Reverse Break Down Voltage. * Identification Of Diode, Led, Resistor. A Capacitor From A Mixed Collection Of Such Items.	» Organic Compounds Containing Halogens, » Organic Compounds Containing Oxygen, » Organic Compounds Containing Nitrogen, Biomolecules » *Principles Related to Practical Chemistry :-* 1. Detection of extra elements (Nitrogen, Sulphur, halogens) in organic compounds. 2. Detection of the hydroxyl (alcoholic and phenolic), carbonyl (aldehyde and ketones), carboxyl and amino functional groups in organic compounds. on of Acetanilide , p-nitro a.	» Organisms and Populations » Ecosystem » Biodiversity and Conservation » Demography » Biology In Human Welfare : Human Health and Disease » Origin And Evolution

Note : Test venue & Time will be displayed on your building's Notice board.

SPACE FOR ROUGH WORK

ALLEN® CAREER INSTITUTE Pvt. Ltd.
Registered & Corporate Office : 'S...r, Kota (Rajasthan) INDIA-324005
Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.in | Website : www.allen.ac.in